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11.8.2 Active Absorption

Absorption of ions against the concentration gradient with the expenditure of metabolic energy is called **active absorption**. In plants, the vacuolar sap shows accumulation of anions and cations against the concentration gradient which cannot be explained by theories of passive absorption. Mechanism of active absorption of salts can be explained through Carrier concept.

Carrier Concept:

This concept was proposed by **Van den Honert** in 1937. The cell membrane is largely impermeable to free ions. However, the presence of **carrier molecules** in the membrane acts as a vehicle to pick up or bind with ions to form **carrier-ion-complex,** which moves across the membrane. On the inner surface of the membrane, this complex breaks apart releasing ions into cell while carrier goes back to the outer surface to pick up fresh ions (Figure 11.25).

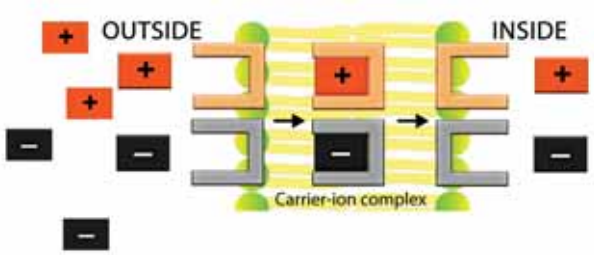


Figure 11.25: Carrier Concept

The concept can be explained using two theories:

1. **Lundegardh's Cytochrome Pump Theory:**

Lundegardh and **Burstrom** (1933) observed a correlation between respiration and anion absorption. When a plant is transferred from water to a salt solution the rate of respiration increases which is called as **anion respiration** or **salt respiration**. Based on this observation **Lundegardh** (1950 and 1954) proposed cytochrome pump theory which is based on the following assumptions:

i. The mechanism of anion and cation absorption are different.

ii. Anions are absorbed through cytochrome chain by an active process, cations are absorbed passively.

iii. An oxygen gradient responsible for oxidation at the outer surface of the membrane and reduction at the inner surface.

According to this theory, the enzyme *dehydrogenase* on inner surface is responsible for the formation of protons (H^+) and electrons (e^-). As electrons pass outward through electron transport chain there is a corresponding inward passage of anions. Anions are picked up by oxidized cytochrome oxidase and are transferred to other members of chain as they transfer the electron to the next component (Figure 11.26).

The theory assumes that cations (C^+) move passively along the electrical gradient created by the accumulation of anions (A^-) at the inner surface of the membrane.

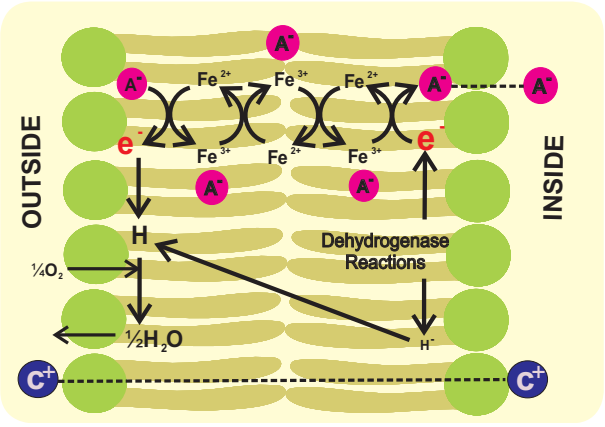


Figure 11.26: Cytochrome Pump theory